RAIR and KAIR Demonstration Script

Presenter names _____

Use this script for a 6 to 8 minute live demonstration after the slides. The examples are deliberately simple so the teacher can inspect both reasoning links. They are demonstrations, not new benchmark questions.

Preflight before entering the classroom

Connect power and keep the local model installed. Keep the report, slides and their PDF copies available offline.

From E:\Projects\role_aware_kair, run .\start_dashboard.cmd. If Ollama is stopped, run ollama serve in a separate terminal first. Open http://127.0.0.1:8765/.

Click the connection indicator and confirm Ollama connected and qwen3.5:4b. Stop here if the model is missing or the service is unavailable. The dashboard does not download models.

Choose Two-hop example · complete evidence. Set base seed 42. Expand Source documents and confirm the two useful sentences below. Allow several minutes for local inference and rehearse once before the presentation.

For an optional automated check, run .\.venv\Scripts\python.exe -X utf8 -B -m unittest discover -s tests. The final release check passed 207 tests. A test pass validates software behavior, not research answer accuracy.

Source	Sentence
The Glass Orchard	The Glass Orchard was written by Mira Solen.
Mira Solen	Mira Solen was born in the city of Nerava.

Opening explanation

Say: We compare two ways of answering a question that needs more than one document. RAIR uses specialized roles and a shared knowledge state. KAIR uses competing agents with private knowledge. Both receive the same question, source pool, model and resource caps.

Say: These names and sources are fictional illustrative inputs. The answer should be Nerava because the book identifies its author and the author document identifies the city. We will run both systems rather than display a prepared answer.

Live demonstration sequence

Time	Action and spoken explanation
0 to 2 min	Show both useful sentences. Click Run comparison once. Explain four retrievals and eighteen model calls per engine. Inputs lock while the run proceeds. A question-free warm-up happens before inference.
2 to 3 min	After completion, inspect both answer panels. In the validated run both returned Nerava. Click a source citation to show original text. Explain that source indexing is checked but an LLM support check can still make mistakes.
3 to 4 min	Open RAIR Execution trace. Show gaps, queries, admitted facts and source references. Open KAIR Execution trace. Show private agent states and the selected proposal. Point out that the agents use the same templates with different seeds.
4 to 6 min	Choose Two-hop example · missing evidence. Show that the Mira Solen source now says only that she writes novels. Run again. Both abstained in validation. Explain that the city cannot be justified from the supplied evidence.
6 to 7 min	Scroll to Actual benchmark results. Explain that this frozen table is separate from the live demo. Show the 22-point gain over the earlier RAIR core and the inconclusive comparison with KAIR.
7 to 8 min	Use Download run JSON for the teacher to inspect the complete record. Close with the main conclusion and one remaining limitation.

Exact result explanation

Say: On 100 fresh questions at two seeds, updated RAIR scored 40.5% exact match and KAIR 36.0%. The paired confidence interval for RAIR minus KAIR is -3 to +12 percentage points, so we cannot claim a conclusive winner. The primary matched control shows a 22-point RAIR core improvement, with an interval of +15 to +29.5 points.

Say: RAIR still abstained in 61 of 200 runs and gave 58 answers that were not exact matches. Some are wording differences, but real reasoning failures remain. Passing our tests does not mean every answer is correct.

Recovery and optional extension

If the live demo does not finish as expected

Keep the result visible and explain its actual status. Do not repeatedly click Run or promise deterministic outputs. Seeds reduce uncontrolled variation but do not guarantee an identical result on every runtime or hardware setup.

Observed issue	Recovery
Ollama unavailable	Check the local service and installed qwen3.5:4b model, then use the connection button. Do not download a large model during the presentation.
Port already in use	Open the existing dashboard, or use .\start_dashboard.cmd --port 8766 and open the corresponding URL.
Long-running comparison	Wait and explain the serial engine order. Closing the browser does not cancel the backend. Stopping the server may interrupt its active engine.
Answer differs or an error occurs	Show its trace honestly. Use the report to explain that verification is fallible. This live case does not change the frozen research result.
No live inference available	Label the fallback as recorded Step 6 evidence. Show docs/assets/dashboard_step6_desktop.png and the saved journal listed below. Do not imply it is a current live run.
Download unavailable after restart	Older results remain as results/dashboard_<run_id>.jsonl on disk, although old in-memory URLs may no longer work.

Saved evidence fallback

Complete case: results/dashboard_86475c0b90134223a03ead59b68c881e.jsonl. Both engines returned Nerava with original source citations. Missing case: results/dashboard_37b889ebd8814f579f6982e2a77383ed.jsonl. Both abstained. Final checks are in results/step7_verification.json.

Optional custom source demonstration

Choose Write your own question and sources. Ask: In which city was the director of The Amber Lantern born? Add The Amber Lantern was directed by Lina Voss, and Lina Voss was born in the city of Selvara, as separate source documents. Seed 43 previously returned Selvara from both engines. Custom cases have no gold score. Do this only if time permits.

Closing sentence: We have completed an inspectable local comparison and demonstrated a clear improvement over the earlier RAIR core. Larger fresh studies and stronger checks are needed before broader claims or real-world reliance.